Software Development Report

An investigation into Lightweight Hand Pose estimation for Edge HCI

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Acknowledgements

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1 Introduction

This project focuses on lightweight 2D RGB hand pose estimation and implements a DRHand-inspired dual-branch approach in a form suitable for training, evaluation, inference, benchmarking, and reproducible experiment reruns.

The software artefact is structured as a research-oriented repository. It combines a reusable Python package for the core hand-pose pipeline with command-line scripts, experiment definitions, preserved checkpoints, and supporting documentation. This report describes the purpose and operation of the software, its design and implementation, the main architectural components and interfaces, the data representations it uses, and the quality-assurance steps taken during development.

2 Purpose, Operation and Validation

The software is intended to implement and evaluate a lightweight 2D hand pose estimation pipeline for edge HCI, with a DRHand-inspired dual-branch model that predicts hand keypoints from RGB images and supports training, inference, benchmarking, and reproducible experiment reruns.

A user operates the software from the command line by setting up the documented Python environment, then running the repository scripts for the required workflow, such as `scripts/train.py` for training, `scripts/eval_metrics.py` for checkpoint evaluation, and `scripts/predict_image.py` for single-image inference, with inputs and options supplied as command-line arguments.

The software takes inputs such as dataset roots, checkpoint files, single RGB images, model and evaluation settings passed through command-line arguments, and it produces outputs such as trained checkpoint files, evaluation JSON metrics, aggregated CSV result tables, benchmark summaries, and qualitative prediction overlays or rendered figures.

A user can tell the software is working correctly when the lightweight test suite passes, the main command-line scripts load and print valid help output, and training, evaluation, inference, or benchmarking workflows execute successfully and produce outputs in the expected format that can be compared against the preserved reference JSON, CSV, checkpoint, and qualitative artefacts bundled in the repository.

3 Design and Implementation

The software is implemented as a modular research artefact whose main design goal is to separate reusable hand-pose functionality from workflow-specific execution logic. Core behaviour is kept in the Python package under `src/handpose`, while command-line scripts under `scripts/` provide thin entry points for training, evaluation, inference, benchmarking, aggregation, and qualitative rendering. This allows the same implementation to support day-to-day experimentation, controlled reruns, and assessment-oriented replication without duplicating logic across multiple workflows.

Within the package, responsibilities are separated by concern rather than by script. The `handpose.data` modules provide a common interface over the supported datasets and keypoint layouts, `handpose.models` defines the DRHand-inspired dual-branch network and its losses, `handpose.training` contains the stage-specific optimisation and artefact logging code, `handpose.inference` implements prediction and branch-fusion behaviour, `handpose.evaluation` computes metrics and result payloads, and `handpose.checkpoints` standardises checkpoint persistence and metadata validation. This structure was intended to

keep the codebase easier to test, extend, and reuse across both the full 21-keypoint configuration and the reduced 10-keypoint experiments.

At implementation level, the system is built around a small set of consistent internal representations: resized RGB inputs, normalized keypoint coordinates, metadata-rich checkpoints, and structured JSON or CSV outputs. Dataset loaders convert dataset-specific annotations into a shared format, the model produces heatmap and coordinate predictions, fused inference combines these branches when required, and the surrounding scripts orchestrate complete workflows such as training, evaluation, benchmarking, and result aggregation. As a result, reproducibility is treated as part of the software design itself rather than as an external add-on, with experiment definitions, preserved checkpoints, and reference artefacts integrated directly into the repository.

3.1 Significant Assumptions Made for the Design and Implementation

Framework equivalence. The software assumes that the DRHand method can be reimplemented in PyTorch even though the original paper describes a TensorFlow implementation. As a result, this repository should be understood as a DRHand-inspired reconstruction rather than a strict reproduction, so exact numerical agreement with the original paper should not be expected.

Dataset substitution. The software assumes that the dual-branch DRHand idea can be studied meaningfully on RHD, HK26K / `coco_hand`, and RH8 even though the original paper used YouTube2D Hands, GANerated Hands, STB, and RHD. This means the software evaluates the method under a different data regime, so its results support the reconstructed pipeline used in this project rather than the original exact benchmark setup.

Prepared input assumption. The software assumes that each input image already contains a clearly visible hand region and therefore does not implement a separate hand detector or crop-normalisation stage. This affects use because users must provide images or datasets where the hand is already reasonably framed, otherwise performance may degrade when the full image is simply resized to the network input size.

Fixed preprocessing strategy. The software assumes that full-image resize to 256 x 256 is an acceptable preprocessing strategy because the paper specifies the input size but does not define a more detailed input-normalisation pipeline. This makes the system simpler and easier to reproduce, but it also means the model is sensitive to changes in hand scale, position, and framing within the input image.

Coordinate representation. The software assumes that keypoints are represented internally as normalized (x, y) coordinates in the resized image space. This affects how the system works because training, inference, fusion, and evaluation all depend on this normalized representation, and conversion back to original pixel coordinates is only done when exporting predictions for human inspection or downstream use.

Heatmap supervision reconstruction. The software assumes Gaussian target heatmaps for heatmap regression because the paper defines an L2 heatmap loss but does not specify exactly how target heatmaps are constructed. This affects optimisation because the heatmap branch is trained to match smooth Gaussian peaks rather than some other target representation, which changes convergence behaviour.

Heatmap width assumption. The baseline system assumes `heatmap_sigma = 2.0` because the original paper does not state a fixed Gaussian width. This matters because the spread of the target heatmap changes the difficulty and smoothness of the heatmap prediction task, so reproducing results depends on keeping this value consistent.

Wing-loss parameter assumption. The software assumes baseline Wing-loss values of w = 10.0 and epsilon = 2.0 because the paper gives the formula but not the actual parameter settings.

This directly affects how the coordinate branch is trained, since different values change the relative penalty for small, medium, and large localisation errors.

Stage-2 optimisation assumption. The software assumes that stage 2 should jointly optimise both the heatmap and coordinate branches after the coordinate branch is added. This means the second training stage is not treated as coordinate-only fine-tuning, and the final model behaviour reflects continued interaction between both branches during optimisation.

Loss-weighting assumption. The baseline implementation assumes equal stage-2 loss weights for heatmap and coordinate supervision because the paper does not specify how the two losses should be balanced. This affects performance because the default behaviour gives both branches equal influence, while changing these weights shifts training emphasis toward one branch or the other.

Parameter update policy. The default implementation assumes that the backbone and heatmap branch remain trainable during stage 2 unless freezing is explicitly requested. This affects how the model learns because the full network can continue adapting in the second stage instead of treating stage-1 features as fixed.

Effective batch-size reconstruction. The software assumes that the paper's stage-2 batch size of 256 can be approximated using batch size 64 with gradient accumulation on different hardware. This makes the training pipeline usable on more limited systems, but it also means the repository reproduces the paper's intended effective batch size only approximately rather than literally.

Convergence assumption. The paper states that stage 1 runs until convergence and mentions 11 epochs in its original setup, but it does not define a complete stopping rule for this implementation context. The software therefore assumes a larger epoch budget together with early stopping, which means training duration is reconstructed pragmatically rather than copied exactly from the paper.

No explicit augmentation policy. The software assumes a simple deterministic preprocessing pipeline because the paper does not specify a detailed augmentation strategy.

Single-hand target assumption. The software assumes one target hand per sample, selecting the requested or most visible hand where needed rather than jointly predicting multiple hands. This affects use because the repository is designed for single-hand pose estimation workflows rather than general multi-hand scene understanding.

Fusion threshold definition. The paper says fusion uses the median predicted knuckle length as the adaptive threshold alpha, but it does not fully formalise the exact skeleton edges in text. The implementation therefore assumes explicit edge lists for supported skeleton layouts, which means the behaviour of fusion depends partly on this code-level reconstruction.

Reduced 10-keypoint extension. The software assumes that the same overall DRHand-style architecture can be reused for a reduced 10-keypoint fingertip-and-base layout even though the original paper is written for 21 keypoints. This allows the project to test whether a smaller output space is sufficient for edge-HCI use, but it also means the reduced model is an adaptation introduced by this project rather than something directly defined in the paper.

Visibility-aware evaluation assumption. The software assumes that evaluation should mask invisible joints when visibility information is available. This affects the reported metrics because performance is measured primarily on visible landmarks, which may differ from alternative evaluation protocols. This is a consequence of the annotation specifics of the RHD dataset. Where often the coordinates of hands outside the image frame are still recorded.

Reduced-layout EPE root assumption. The reduced shared-10 layout does not include the same root structure as the full 21-keypoint skeleton, so the implementation assumes an

alternative root for root-relative EPE when necessary. This keeps EPE usable for reduced-layout experiments, but it means that reduced-layout EPE is not a perfectly identical measurement setup to the full-layout case.

3.2 Languages, Packages, Frameworks, and Libraries Used

Python. The software was built primarily in Python because it provides a practical and widely used language for machine-learning research, rapid experimentation, data processing, and command-line workflow development. This made it suitable for implementing a research artefact that needed to support training, evaluation, inference, benchmarking, and reproducibility.

PyTorch. The main hand-pose pipeline was implemented in PyTorch because the project required a flexible deep-learning framework for defining custom model architectures, losses, and training procedures. PyTorch was particularly suitable for this DRHand-inspired implementation because it allowed the original dual-branch idea to be reconstructed clearly while remaining easy to adapt for the reduced 10-keypoint and transfer-learning experiments.

Conda. A Conda-managed environment was used because the project depended on a mix of machine-learning, image-processing, and benchmarking packages that needed to be installed consistently across systems. Conda was chosen to reduce dependency conflicts and to make the installation and replication process more reliable for assessment.

NumPy. NumPy was used for numerical array processing because the software required lightweight handling of coordinates, image-derived data, and intermediate structured values outside the main tensor-training code. It was a natural choice because it integrates well with Python-based scientific and machine-learning workflows.

Pillow. Pillow was used for image loading and resizing because the project processes RGB hand images during training, evaluation, and inference. It was chosen as a simple and dependable library for image I/O and preprocessing.

MediaPipe. MediaPipe was used because the project included an explicit baseline comparison against a strong lightweight hand-tracking system and also relied on MediaPipe-related assets for pseudo-labelled or baseline-oriented workflows. This choice was important because it gave the project a practical reference point against a widely used real-time hand-pose system.

Matplotlib. Matplotlib was used for plotting and qualitative rendering because the project required visual inspection of predictions, generation of qualitative examples, and support for report-ready analysis figures. It was chosen because it is a standard and effective library for producing these outputs in Python.

PyTest. PyTest was used for automated testing because the repository needed a lightweight way to verify important behaviour such as model output shapes, fusion logic, and checkpoint validation. It was chosen to support software quality assurance and to provide a quick validation route for assessors and future users of the artefact.

psutil. The `psutil` package was used in benchmarking workflows because the project included deployment-oriented runtime analysis and needed process-level resource measurements in addition to model accuracy. It was chosen to support practical evaluation of the software on edge-oriented hardware settings.

3.3 Overall Architecture / System Design and Interfaces Between Components

Overall Structure

Figure 1 shows the overall system structure. The repository is designed as a modular research artefact in which thin command-line scripts provide the user-facing workflows, while the reusable

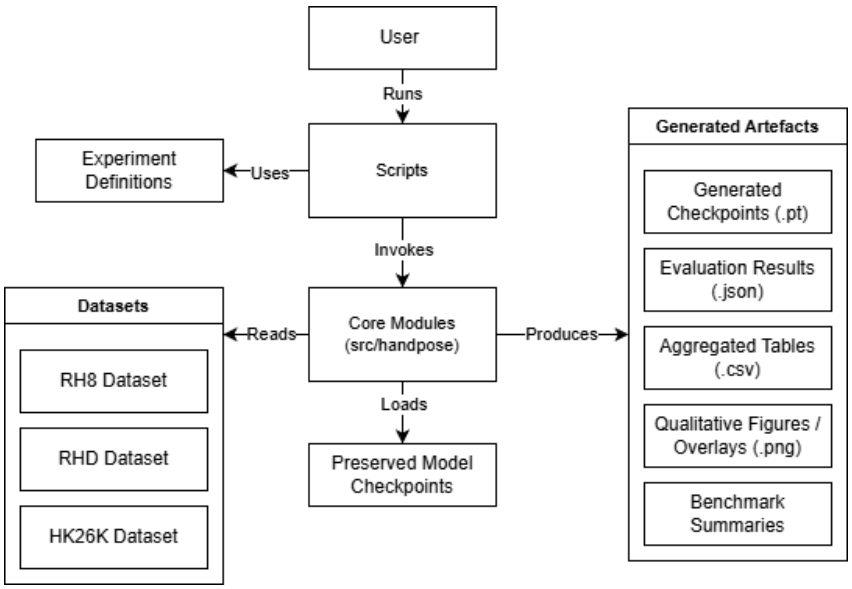

Figure 1: High-level architecture.

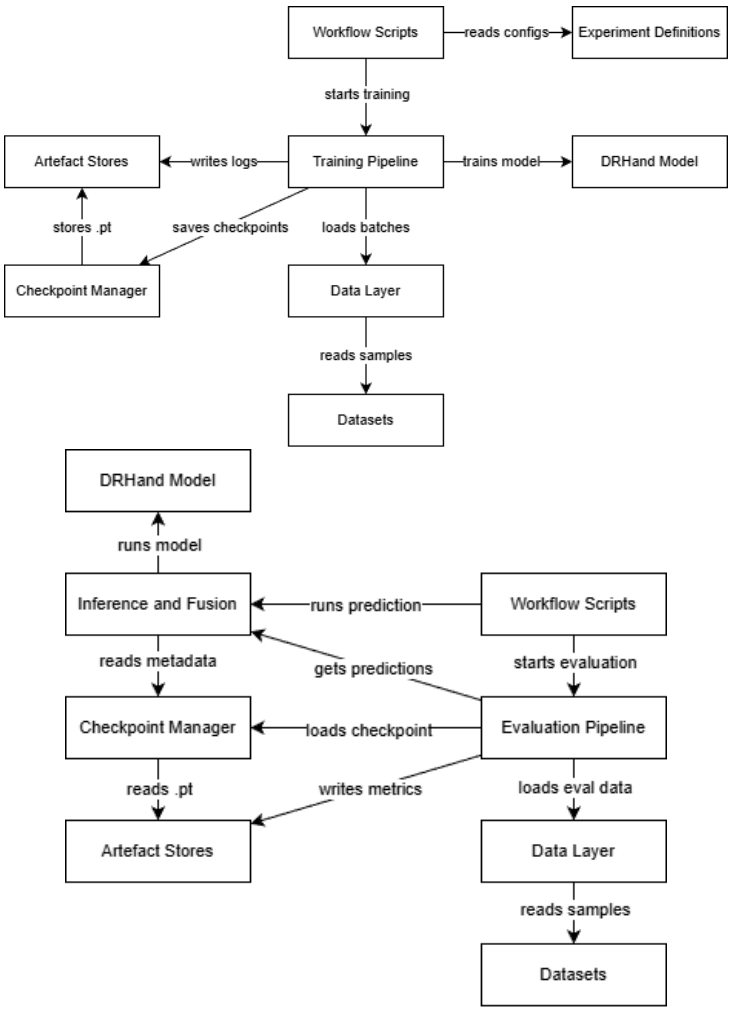

Figure 2: Training and evaluation component views.

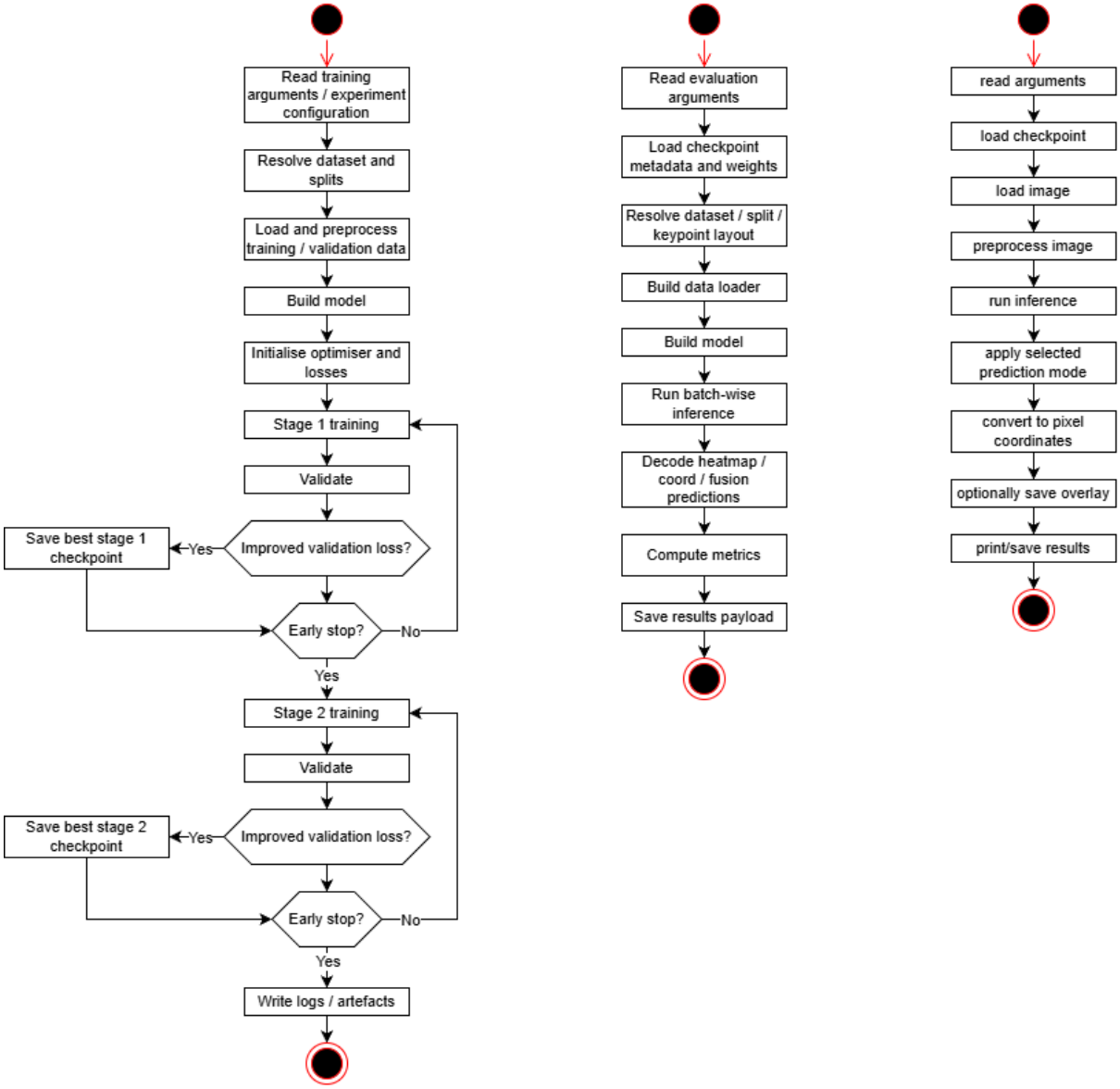

Figure 3: Training, evaluation and inference activity diagrams.

implementation is contained in the `src/handpose/` package. This separates workflow orchestration from the core hand-pose logic and allows the same internal code to support training, evaluation, benchmarking, and single-image inference.

At the highest level, the user interacts with the software by running scripts such as `scripts/train.py`, `scripts/eval_metrics.py`, and `scripts/predict_image.py`. These scripts optionally use experiment definitions and generated command files, then invoke the core modules responsible for data loading, model execution, training, inference, evaluation, and checkpoint handling.

The core modules read dataset files from disk and produce new artefacts such as trained checkpoints, evaluation JSON files, aggregated CSV tables, benchmark summaries, and qualitative figures. This structure keeps the execution interface simple while ensuring that the main implementation remains reusable and consistent across workflows.

Training View

Figure 2 expands this design into separate training and evaluation component views. In the training view, the workflow scripts start the training pipeline, which requests batches from the data layer, passes them through the DRHand model, computes the required losses, and updates the model parameters.

The data layer provides a shared interface for loading and preprocessing supported datasets into normalized tensor representations. During training, the checkpoint manager validates and saves model state together with metadata, while the training pipeline also writes logs and loss histories to the artefact store.

Evaluation and Inference View

In the evaluation and inference view, the workflow scripts start either checkpoint evaluation or direct prediction. The evaluation pipeline loads a saved checkpoint through the checkpoint manager, rebuilds the model configuration, obtains evaluation samples from the data layer, and requests predictions from the inference and fusion module.

That module runs the DRHand model, decodes heatmap outputs, uses the coordinate branch predictions, and applies the fusion rule when fused inference is requested.

The resulting predictions are then used by the evaluation pipeline to compute metrics such as SSE, EPE, PCK, timing information, and optional fusion diagnostics, which are written back to the artefact store as structured outputs.

Workflow Activities

Figure 3 complements the component views by showing the workflows as activities. It emphasizes the execution order of the main repository tasks: training moves from dataset loading and preprocessing to staged optimisation, validation, and checkpoint export, while evaluation and inference move from checkpoint loading and sample preparation to prediction, fusion, metric computation, and output generation.

Interfaces Between Components

The interfaces between these components are mainly file-based and tensor-based. File-based interfaces include dataset roots, annotation files, checkpoint `.pt` files, evaluation `.json` outputs, summary `.csv` files, and rendered `.png` figures.

In-memory interfaces are primarily PyTorch tensors representing images, keypoint coordinates, visibility values, heatmaps, and predicted coordinates. Checkpoint metadata provides an

additional interface between workflows by preserving information such as training stage, epoch, keypoint layout, and other configuration details needed to reload and evaluate a model correctly.

3.4 Data Model(s), Including In-Memory Structures and Persistent Storage

Processed Data

The main inputs are RGB hand images and per-image annotations from the supported datasets, including 2D keypoint coordinates and visibility values. During preprocessing, these inputs are converted into a shared tensor representation that is reused across training, evaluation, and inference.

In-Memory Representation

The main in-memory structures are image tensors of shape `(C,H,W)`, keypoint-coordinate tensors of shape `(K,2)`, visibility tensors of shape `(K)`, predicted heatmaps of shape `(K,64,64)`, and predicted coordinate tensors of shape `(K,2)`.

Persistent Storage

Persistent storage in the repository is file-based. Source datasets remain on disk as image and annotation files, while intermediate and final outputs are saved as experiment artefacts. Training checkpoints are written as PyTorch `.pt` files containing model state and metadata, loss histories are stored as CSV, evaluation outputs are written as JSON, aggregated summaries are stored as CSV, and qualitative overlays or rendered figures are exported as `.png`. This design supports reproducibility because trained states, metrics, and generated figures can all be reloaded or compared directly from the filesystem.

Data item	**In-memory form**	**Persistent form**
RGB input image	Preprocessed tensor `(3,H,W)`	`.png` or `.jpg` image files
Ground-truth hand pose	Keypoint tensor `(K,2)` plus visibility tensor `(K)`	RHD annotations or COCO-style `.json` annotations
Training / evaluation batch	Batched tensors such as `(N,3,H,W)` and `(N,K,2)`	Runtime only; constructed by data loaders
Model predictions	Heatmaps `(N,K,64,64)` and coordinates `(N,K,2)`	Exported prediction or metric payloads in `.json`; optional qualitative `.png` outputs
Checkpoint state	Python dict containing `model_state`, keypoint layout, stage, epoch, and training metadata	PyTorch checkpoint `.pt` files
Result summaries	Scalar metrics and aggregated experiment statistics	CSV tables and JSON result files

Table 1: Main data representations used by the repository.

3.5 Significant Methods, Functions, or Other Lower-Level Elements

Core Model

The most important lower-level implementation element is `HandPoseNet` in `src/handpose/models/hand_pose_model.py`, which acts as the main wrapper around the learned model.

It combines the shared backbone, heatmap regression head, and coordinate regression head into one interface and provides the forward paths used during both training and inference. This makes it the central model component on which the rest of the repository depends.

Architecture and Optimisation

At the architectural level, the most significant lower-level classes are `Backbone`, `Heatmap_reg`, and `coord_reg` in `src/handpose/models/architecture.py`, together with the reusable block definitions in `src/handpose/models/block.py`.

These classes implement the shared feature extractor, the heatmap branch, the coordinate branch, and the depthwise-separable and upsampling building blocks from which the DRHand-inspired model is assembled.

The most important optimisation-related functions are `coords_to_heatmaps`, `HeatmapMSELoss`, and `WingLoss` in `src/handpose/models/losses.py`, along with `build_optimizer_stage1`, `build_optimizer_stage2`, `train_one_epoch`, and `validate` in the training modules.

These functions define how supervision targets are constructed, how the two training losses are computed, how different stages of the model are optimised, and how batch-wise training and validation are carried out.

Data Processing and Fusion

Another important group of lower-level functions appears in the preprocessing and dataset-loading code. The dataset factory and dataset classes in `src/handpose/data/` provide a unified interface for loading RHD and COCO-style hand datasets, while preprocessing helpers in `src/handpose/data/transforms.py` resize images, convert them to tensors, and rescale keypoint coordinates. These functions are significant because all training, evaluation, and inference workflows depend on the same shared input format.

The lower-level functions implementing branch fusion. In `src/handpose/inference/fusion.py`, functions such as `heatmaps_to_coords_argmax`, `resolve_fusion_bone_edges`, and `fuse_coords` define how heatmap predictions are decoded and how the final fused prediction is selected from the heatmap and coordinate branches. These functions are central because the fusion rule is one of the defining design features of the implemented pipeline.

Checkpointing and Workflow Entry Points

Checkpoint and evaluation handling. Functions in `src/handpose/checkpoints.py` define how checkpoints are validated, saved, and reloaded, while `evaluate_checkpoint` in `src/handpose/evaluation/eval_metrics_core.py` performs the main metric computation for SSE, EPE, PCK, timing, and optional fusion diagnostics. These components are important because they make the repository reproducible and allow trained models to be reused consistently across training, inference, and result generation workflows.

The user-facing scripts under `scripts/` are thin but important workflow connectors. Scripts such as `train.py`, `eval_metrics.py`, `predict_image.py`, and `benchmark_pipeline.py` do not contain the main modelling logic themselves, but they are the main entry points through which a user accesses the lower-level implementation for complete training, evaluation, inference, and benchmarking tasks.

3.6 Important Errors and Their Conditions

Input and Configuration Errors

Several important errors in the software are caused by invalid inputs, missing files, or inconsistent configuration between artefacts. For example, the checkpoint utilities raise errors when a checkpoint is missing required metadata fields, has a mismatched keypoint layout, contains duplicate keypoint indices, or uses an unsupported checkpoint version. Similar validation errors occur when invalid keypoint subsets are requested, when unsupported dataset names or split aliases are supplied, or when an invalid prediction mode or negative PCK threshold is passed on the command line. In these cases, the software responds by raising a clear `ValueError` and stopping execution rather than continuing with inconsistent state.

Runtime Resource Errors

A second group of important errors occurs when required runtime resources are not available. Examples include missing dataset directories, missing image files, empty evaluation image folders, failed imports of optional dependencies such as MediaPipe, or requesting CUDA on a machine where it is not available. These conditions are handled through exceptions such as `FileNotFoundError` or `RuntimeError`. The software does not silently ignore these problems; instead, it fails early and reports the missing dependency, missing path, or unsupported device condition so that the user can correct the environment before rerunning the workflow.

Workflow State Errors

Further errors can arise from invalid workflow state during training or evaluation. For example, stage-specific training functions raise errors if an unsupported stage number is used, if gradient accumulation is configured with an invalid value, or if stage 2 is requested without the required coordinate loss. Training also raises an error if `-skip-stage1` is used without providing an initial checkpoint.

Evaluation and qualitative-rendering workflows can also fail when no samples match the requested dataset split or when the selected checkpoint does not contain the keypoints required for shared-10 evaluation or fusion. In these situations, the software again responds by terminating with an explicit error message, which helps prevent silent misuse of the repository and makes debugging more straightforward.

3.7 User Interface Design

Command-Line Interaction

The user interacts with the software primarily through a command-line interface. The main workflows are exposed as separate Python scripts under `scripts/`, including training, evaluation, benchmarking, single-image inference, qualitative rendering, and result aggregation.

In practice, the user runs these scripts from the repository root, supplies the required paths and options as command-line arguments, and receives outputs such as checkpoints, JSON result files, CSV summaries, printed metrics, or rendered visualisations.

This command-line design was chosen because the software is a research artefact rather than an end-user application. The main tasks involve running controlled experiments, reproducing reported results, evaluating checkpoints under different conditions, and generating analysis artefacts.

These activities are better supported by explicit script arguments than by a graphical user interface, because command-line workflows are easier to document, automate, reproduce, and integrate with local machines, Conda environments, and HPC or Slurm-based execution.

Separation of Interface and Core Logic

Another important interface choice was to separate the reusable implementation from the user-facing entry points. The core logic is implemented in the `src/handpose/` package, while the scripts act as thin workflow wrappers around that logic. This was done so that the user interface remains simple at the script level while the lower-level code remains modular and reusable. As a result, users can run complete workflows through a small number of documented commands without needing to interact directly with the internal package structure.

Explicit Runtime Configuration

The interface relies on explicit file and path arguments for datasets, checkpoints, and output locations. This choice was made to keep the repository portable across different machines and storage layouts, especially since some datasets are external and not bundled with the artefact. By making these dependencies explicit at runtime, the software avoids hard-coding machine-specific paths and makes the execution process clearer for assessors and future users.

3.8 Quality Assurance Steps

Modular Structure

Several steps were taken to improve software quality throughout the project. The repository was organised into a modular structure, with reusable implementation code placed under `src/handpose/` and user-facing workflows exposed through separate scripts. This separation reduced duplication, made the system easier to understand, and improved maintainability by ensuring that training, evaluation, inference, and benchmarking all relied on shared core logic rather than separate ad hoc implementations.

Validation and Error Handling

Software quality was also improved through explicit validation and defensive error handling. Important interfaces such as checkpoint loading, dataset selection, prediction-mode resolution, keypoint-layout handling, and preprocessing inputs include checks that raise clear exceptions when invalid arguments or inconsistent artefacts are supplied. This helped reduce silent failure modes and made incorrect usage easier to identify and debug.

Testing and Reproducibility

Correctness and reliability were checked using the lightweight automated test suite under `tests/`. These tests verify important repository contracts such as model output shapes, checkpoint schema validation, and fusion behaviour. In addition to automated tests, the repository also provides practical validation routes through documented commands in `INSTALL.md` and `REPLICATION_GUIDE.md`, allowing the software to be checked by running real workflows and comparing regenerated outputs against preserved reference artefacts.

Documentation and Artefact Management

Maintainability was further supported through reproducibility-oriented documentation and artefact management. The repository includes installation instructions, requirement notes, a replication guide, curated experiment command files, preserved checkpoints, and saved result artefacts. These were important quality-assurance measures because they made the software easier to inspect, rerun, and extend, while also ensuring that the reported experimental outputs could be traced back to concrete workflows within the repository.

3.8.1 Version Control and Versioning

Version control was used throughout development by maintaining the project as a Git-based repository. This allowed the software, documentation, experiment definitions, and preserved artefacts to be developed within a single structured history rather than as disconnected files. Using version control was important because the project involved repeated changes to model design, training configuration, evaluation workflows, reporting artefacts, and supporting documentation, all of which needed to remain traceable over time.

Changes were tracked and managed through the normal Git workflow of recording edits to the repository as development progressed. This made it possible to monitor the evolution of scripts, source modules, experiment files, and paper materials, and helped prevent important changes from being lost or mixed together without record. It also supported safer experimentation, since alternative implementations or revised configurations could be introduced while preserving the earlier state of the project.

3.8.2 Code Style and Convention

The code follows a consistent Python style based on clear module separation, descriptive function names, and lightweight internal documentation. The repository is organised by responsibility, with separate modules for data loading, model definition, training, inference, evaluation, and checkpoint handling. Within these modules, functions and classes generally use descriptive snake_case naming for functions and variables, and class-based naming for model components such as `HandPoseNet`. This made the code easier to navigate because related logic was grouped together rather than scattered across large scripts.

A consistent convention was also followed for docstrings and purpose statements. Most modules begin with a short summary docstring explaining their role in the repository, and most important functions include concise docstrings describing what they do. This helped keep the code clear because users and future readers can understand the role of a file or function without having to infer it entirely from implementation details.

The code also follows a convention of keeping user-facing workflow logic separate from reusable implementation logic. Scripts under `scripts/` mainly parse arguments and call functions from the `src/handpose/` package, while the package itself contains the lower-level logic. This convention improved consistency because training, evaluation, inference, and benchmarking were all built on shared internal code rather than separate duplicated implementations.

The use of explicit validation and informative error messages at key interfaces. Functions that depend on specific assumptions, such as checkpoint schema, prediction modes, dataset names, keypoint layouts, or device availability, usually check these conditions directly and raise clear exceptions if they are violated. This kept the behaviour of the software more predictable and made the codebase easier to maintain, since failures tended to occur close to the source of the problem rather than emerging later as ambiguous downstream bugs.

The repository shows a consistent testing convention in which small focused tests are used to protect important contracts such as model output shapes, checkpoint validation, and fusion behaviour. This helped reinforce code clarity and consistency because the expected behaviour of core components was made explicit both in implementation and in accompanying tests.

3.8.3 Automated Testing

The repository includes a lightweight automated test suite implemented with `pytest`. The tests are intended as a fast validation layer that checks a small number of important software contracts before running more expensive workflows such as full training or dataset-based evaluation. This

makes them useful both during development and as a first confidence check for assessors or future users of the repository.

One group of tests verifies model-shape behaviour. In `tests/test_model_shapes.py`, the test suite checks that `HandPoseNet` produces outputs of the expected shape for both the standard 21-keypoint configuration and the reduced 10-keypoint configuration. These tests verify that the model returns heatmaps of shape `(N, K, 64, 64)` and coordinate outputs of shape `(N, K, 2)`, and also confirm that these outputs remain finite. This protects the most basic interface contract of the learned model.

A second group of tests verifies the fusion-related helper functions. In `tests/test_fusion.py`, the software checks that heatmap argmax decoding returns the expected normalized coordinates, that the standard 21-joint fusion edge definition is resolved correctly, and that the reduced tip-and-base 10-joint layout is handled correctly while unsupported layouts are rejected. These tests verify the correctness of important lower-level logic used in fused inference.

A third group of tests verifies checkpoint schema handling. In `tests/test_checkpoints.py`, the suite checks that valid checkpoints are accepted and normalized correctly, while malformed checkpoints with duplicate keypoint indices or inconsistent keypoint metadata are rejected. These tests verify the repository's checkpoint contract, which is important because the same checkpoint format is used across training, evaluation, inference, and preserved artefact reuse.

Finally, the `tests/conftest.py` configuration ensures that the `src/` tree is importable when tests are run from the repository root. This supports the automated test workflow by making the repository package available consistently during test execution.

4 Interactive Demo

4.1 Purpose and Scope

Alongside the main hand-pose research artefact, a separate gesture-driven demonstrator was also developed for live presentation. This software is not part of the core `src/handpose/` repository implementation described in the earlier sections. Instead, it should be understood as a companion application whose purpose is to present selected project-related image processing outputs through direct hand interaction rather than through command-line experiments.

This distinction is important because the two software artefacts serve different roles. The main system is a reproducible research implementation for training, evaluation, benchmarking, and inference, whereas the demonstrator is a front-end showcase intended for interactive use in a live setting.

4.2 Operation and User Workflow

The demonstrator starts by opening a ZED camera, creating a fullscreen portrait PySide6 window, and waiting for a hand to be detected. Once a hand is visible, the palm centre is used to control an on-screen cursor and a fist gesture is interpreted as a click.

The user workflow is deliberately simple. A start screen leads to a menu, from which the user can select one of two interactive demonstrations: a segmentation view or a tone-mapping view. Navigation between pages is carried primarily through hand movement and fist-based selection, so the system can be used during presentation without relying on a conventional mouse or keyboard even though the underlying Qt widgets still retain their normal click behaviour.

The interaction logic is implemented explicitly rather than through a generic gesture library. The cursor position is derived from the mean position of a small set of palm-related landmarks,

mirrored horizontally to match user expectation, and then smoothed before being displayed so that motion appears stable rather than jittery. A fist is detected by comparing fingertip-to-base distances against palm width, and clicks are debounced so that one closing motion does not trigger repeated activations. If no hand is detected for a short timeout period, the interface automatically returns to the start screen, which helps recover from tracking loss during live use. In practical terms, this gives the demonstrator two forms of interaction: continuous hand motion is used to control cursor-like behaviour, while discrete fist closures are used as selection events for navigation and page-specific actions.

4.3 Architecture and Main Components

The demonstrator can be understood as a small interactive architecture with three main parts, as shown in Figure 4. First, a perception pipeline acquires live frames from the ZED camera, rotates and preprocesses them, and applies MediaPipe Hands to estimate landmarks from which cursor position, palm width, and fist-click state are derived. Second, a central `MainWindow` acts as the controller for the application: it receives pose updates, maintains the active page, updates the overlay, routes gesture-derived click events to the appropriate page logic, and triggers timeout recovery when tracking is lost. Third, the presentation layer is implemented as a stacked set of pages for the start screen, menu, segmentation demonstration, and tone-mapping demonstration, with the content pages loading prepared visual assets from disk.

This structure is an important design choice because it separates the latency-sensitive perception path from the presentation layer while still allowing both demonstrations to share the same interaction controller. Continuous hand motion drives cursor movement and the tone-mapping spotlight, whereas fist closures are treated as click events for page navigation and segmentation selection. Camera acquisition and hand-tracking would make the interface feel unresponsive if they were executed directly in the GUI event loop, so the vision thread emits pose updates to the main window through Qt signals while a timer refreshes the overlay at a steady rate. This reduces the risk of the interface freezing during active use and makes the flow of interaction state easier to reason about.

4.4 Demonstration Behaviour

The first demonstration is event-driven. It presents microscopy imagery together with precomputed segmentation data, and a user selection event determines which prepared overlay is revealed. In one mode, a binary mask is converted into connected components and the selected component is highlighted when the user clicks on the image. In another mode, multiple precomputed RNvU retina masks are loaded and the mask covering the selected pixel is revealed.

The second demonstration is continuous rather than event-driven. It presents tone-mapping results using paired images: a baseline image is shown across the whole display, while an improved image is revealed only inside a hand-controlled spotlight. The spotlight position follows the gesture-controlled cursor and its radius scales with palm width, so hand motion acts as a live control input rather than as a sequence of clicks. In both demonstrations, the software does not run segmentation or tone-mapping models live. It instead provides an interaction layer for exploring prepared outputs visually.

The data handling for these demonstrations is therefore simple but explicit. In the single-mask segmentation mode, the software loads one base microscopy image and one binary mask, computes connected components from the foreground pixels, and stores pre-rendered coloured overlays for each selectable region. In the RNvU mode, the software instead loads a base image together with several binary class masks and combines them into an integer label map so that a click can be resolved to a specific tissue class. The tone-mapping view uses fixed pairs of baseline

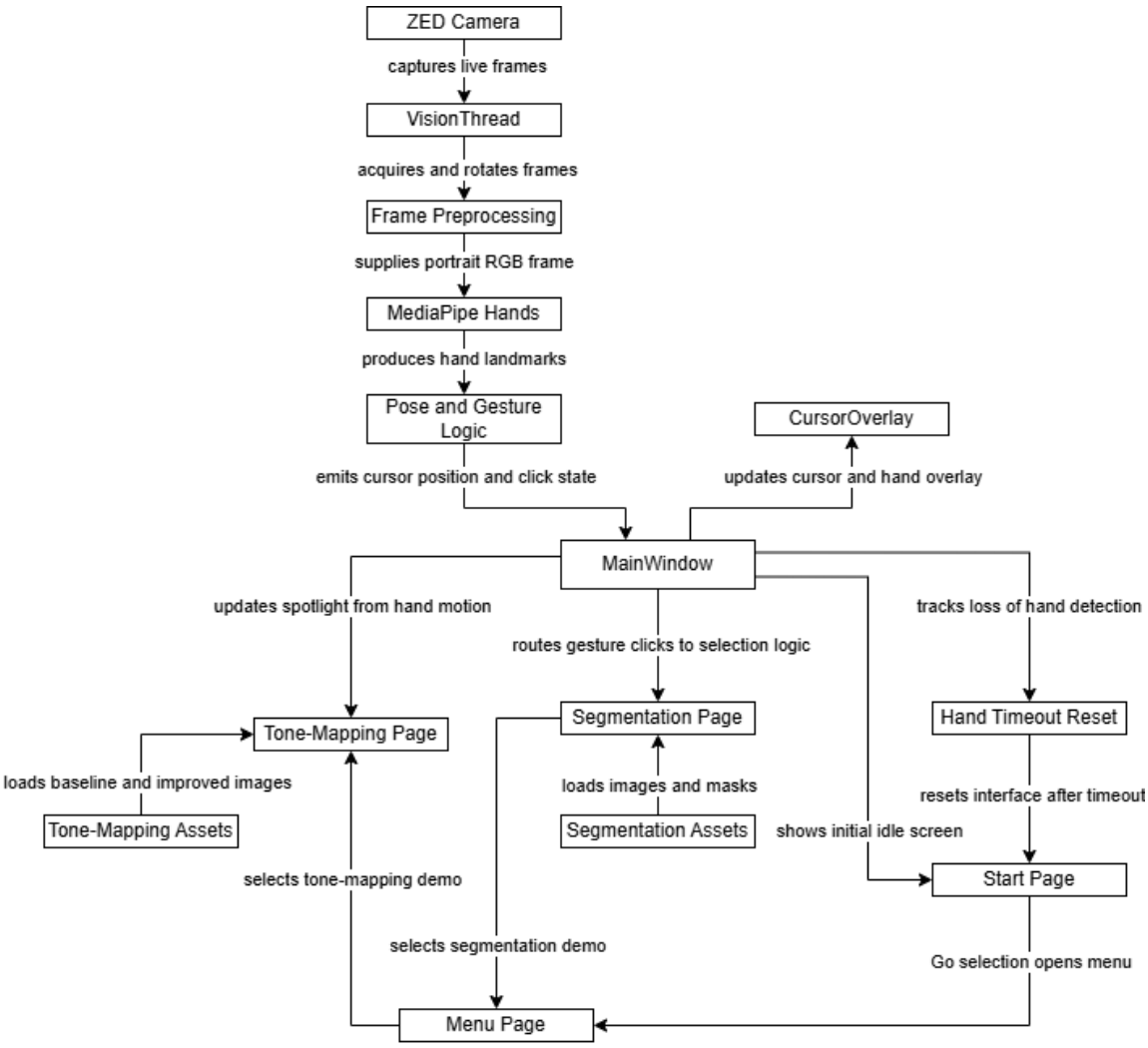

Figure 4: Architecture of the interactive demonstrator, showing the live hand-tracking pipeline, central window controller, and gesture-driven demo pages.

and improved images, and the size of the spotlight is derived from the tracked palm width so that the reveal area scales with the user's hand.

4.5 Inputs, Outputs, Dependencies, and Limitations

The main runtime inputs combine live camera frames from the ZED device with local image, mask, and branding assets prepared for the demonstration. The main software dependencies are Python 3.10, PySide6, OpenCV, MediaPipe, `pyzed`, Pillow, and NumPy.

Several simplifying assumptions were made for this demonstrator. It assumes that a ZED camera is connected, that a single hand can be tracked reliably enough for interaction, and that the application will be used in a fixed portrait kiosk-style layout. It is also only loosely coupled to the main project implementation, because it does not call the DRHand-inspired training or inference pipeline directly and instead visualises precomputed segmentation and tone-mapping assets.

Some failure conditions are also easy to identify from the implementation. The software may fail to start if the ZED camera cannot be opened, and parts of the interface may degrade if required image or mask files are missing, unreadable, or empty. Because the demonstration assets are referenced through relative paths, the application also depends on being launched from the expected working directory. These are ordinary engineering limitations for a presentation tool, but they are worth documenting because they affect deployability more directly than the main research pipeline does.

4.6 Practical Validation

Validation of the demonstrator is practical rather than benchmark-oriented. The software can be considered to be operating correctly when it starts successfully, detects the camera, shows stable cursor motion in response to hand movement, registers fist gestures as clicks with acceptable reliability, switches between pages correctly, highlights the intended segmentation regions, and updates the tone-mapping spotlight in real time.

A useful manual acceptance route is to verify the demonstrator in stages: first, confirm that startup succeeds and the camera is detected; second, confirm that presenting one hand reveals the cursor and allows the menu to be selected; third, confirm that a fist activates buttons only once per intended selection; fourth, confirm that segmentation clicks reveal the expected region or RNvU mask; fifth, confirm that the tone-mapping spotlight follows the hand and changes size with palm width; and finally, confirm that removing the hand causes the interface to reset after the configured timeout. This sequence gives a more precise notion of correctness than a simple statement that the application "works".

A Appendices

A.1 Supervisor Meeting Minutes

The following minutes are presented in concise form and record a brief summary of each meeting, the matters discussed, and the agreed actions.

2025-10-08

Summary: Initial project-scoping meeting. Discussed: lightweight hand pose tracking for real-time HCI on edge hardware, possible use of RGB and stereo input, and the need for an early literature review. Actions agreed: begin literature review, identify core papers, and prepare an initial problem statement.

2025-10-14

Summary: Follow-up meeting on early literature findings. Discussed: latency versus accuracy on edge devices, candidate hardware, and how pose quality would relate to HCI usability. Actions agreed: continue literature review, shortlist likely baselines and datasets, and start outlining the preliminary article.

2025-10-21

Summary: Reported progress on an exploratory hand-tracking mini project. Discussed: using a ZED camera and MediaPipe Hands to gain practical experience with hand-tracking behaviour, including lighting, range, and gesture stability. Actions agreed: continue the exploratory demo as hands-on domain familiarisation while keeping it separate from the main research contribution.

2025-10-28

Summary: Reviewed exploratory demo progress and article direction. Discussed: practical constraints observed in camera-based hand interaction and how these could inform the motivation for the project. Actions agreed: draft the introduction and motivation sections and continue the structured literature review.

2025-11-04

Summary: Literature review review meeting. Discussed: MediaPipe Hands, lightweight hand-pose approaches, stereo or depth-related methods, and how to position the project around edge HCI rather than model accuracy alone. Actions agreed: deepen the review, organise references, and continue drafting the preliminary article.

2025-11-11

Summary: Discussed the developing preliminary article. Discussed: project framing around lightweight hand pose tracking for edge devices, likely hardware and sensing setup, and the need to state clear objectives. Actions agreed: refine the abstract and research aim and improve the structure of the literature review.

2025-11-18

Summary: Progress review before the preliminary submission. Discussed: draft introduction and literature review, possible evaluation themes such as latency, stability, and usability, and how much preliminary implementation to mention. Actions agreed: complete a near-final draft of the preliminary article and prepare the progress appendix.

2025-12-02

Summary: Final pre-submission review before the preliminary article deadline. Discussed: article structure, wording of the objectives, literature coverage, and the role of the exploratory demo as early practical work. Actions agreed: make final edits, proofread, and submit the preliminary research article by 2025-12-05.

2025-12-15

Summary: Post-submission review after the preliminary article. Discussed: lessons from the initial phase, narrowing the project to a more deliverable software artefact, and possible next implementation steps. Actions agreed: define a clearer implementation plan and prepare for the main development phase over the break.

2026-01-06

Summary: Restart meeting after the winter break. Discussed: refining scope from a broad stereo/RGB investigation toward a more manageable core system with reproducible software and evaluation. Actions agreed: define the main implementation scope, organise the repository, and begin the next development phase.

2026-01-13

Summary: Reviewed software architecture direction. Discussed: reusable package structure, scripts for training, evaluation and inference, likely datasets, and the importance of a strong baseline. Actions agreed: implement the core project structure and begin assembling the training and evaluation workflow.

2026-01-20

Summary: Reported progress on codebase setup. Discussed: package organisation, dataset handling, checkpointing, and the need to keep the system reproducible and well documented from an early stage. Actions agreed: continue the core implementation and validate the basic training and inference path.

2026-02-03

Summary: Model-development review meeting. Discussed: adopting a DRHand-inspired direction, staged training, loss design, and balancing technical ambition with a feasible implementation. Actions agreed: proceed with the selected architecture, document assumptions, and begin controlled experiments.

2026-02-17

Summary: Reviewed early implementation and evaluation progress. Discussed: training behaviour, checkpoint management, evaluation scripts, and the importance of preserving reproducible outputs for the final reports. Actions agreed: continue experiments, improve workflow robustness, and preserve representative outputs.

2026-02-24

Summary: Intermediate results review. Discussed: practical performance of the current pipeline, comparison against baseline ideas, and how to connect the software work back to the edge-HCI motivation. Actions agreed: continue evaluation, improve result organisation, and begin outlining the final research and software reports.

2026-03-03

Summary: Focus on implementation maturity and supporting infrastructure. Discussed: evaluation outputs, benchmarking, reproducibility, and the role of supporting software beyond the core model. Actions agreed: improve scripts, artefact management, and documentation, and prepare report-ready material.

2026-03-10

Summary: Shifted attention toward the Software Development Report. Discussed: report structure, architecture description, quality-assurance material, and what evidence should be included for testing and documentation. Actions agreed: draft software report sections and ensure the repository documentation supports installation and reuse.

2026-03-16

Summary: Reviewed final-report and presentation preparation. Discussed: research article drafting, software report content, presentation planning, and how the earlier interactive demo could be described as a companion exploratory artefact. Actions agreed: continue writing, refine the software documentation, and prepare material suitable for the final demonstration and slides.

2026-03-25

Summary: Final major supervision meeting before the submission run-in. Discussed: remaining writing tasks, repository documentation, appendix material, and preparation for the 2026-04-27 deadlines for the research article, software development report, and presentation slides. Actions agreed: complete final experiments and writing, finalise the repository documentation set, prepare the presentation, and assemble the appendices for submission.